Rates of Change and Rectilinear Motion

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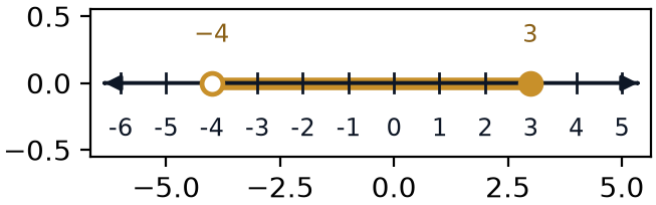
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Position, Velocity & Speed

A particle's velocity is given by $v(t) = 3t^2 - 12t + 9$ for $t \geq 0$. The position of the particle at time $t = 0$ is $s(0) = 5$. \\ Consider a second particle whose position $p(t)$ is described by the integral $p(t) = \int_0^t v(u)\,du$. \\ Find the net displacement of the first particle from $t = 1$ to $t = 3$.

2 [Foundational] • Target: Total Distance vs. Displacement

A particle's velocity is given by the function $v(t) = 5$ for $0 \leq t \leq 4$. Find the total displacement of the particle over this time interval.



3 [Foundational] • Target: Related Rates: Geometric Models

A particle's position in the first quadrant is described by a function $y(t)$, where t is time in seconds. The rate of change of its y-position with respect to time is governed by the differential equation $\frac{dy}{dt} = k\sin(t)$ for some positive constant k. The particle starts at the origin $(0, 0)$ at $t = 0$. At $t = \frac{\pi}{2}$ seconds, its y-position is $y(\frac{\pi}{2}) = 5$. \\ \\ Consider a second particle whose y-position $z(t)$ is described by the differential equation $\frac{dz}{dt} = k\cos(t)$, also starting at the origin at $t = 0$. If the first particle's y-position at $t = \frac{\pi}{4}$ is $y(\frac{\pi}{4}) = \frac{5}{\sqrt{2}}$, what is the value of k? Show your complete algebraic work.

4 [Foundational] • Target: Related Rates: Angles & Proportions

A differentiable function $f(t)$ is represented by the graph below. What is the total change in $f(t)$ from $t = 2$ to $t = 5$? Show your complete algebraic work.

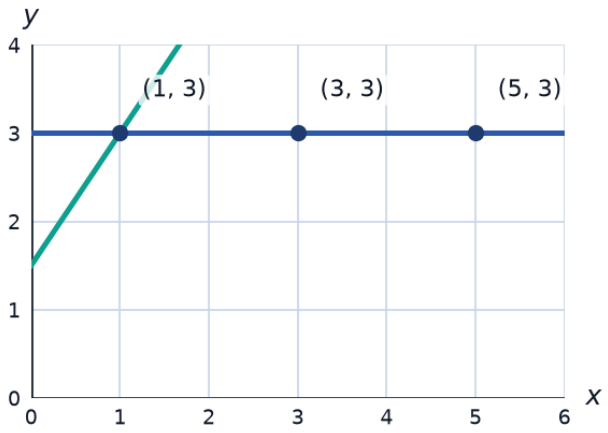
5 [Foundational] • Target: Position, Velocity & Speed

A particle moves with velocity $v(t) = 3t^2$. Displacement from $t = 0$ to $t = 2$ is:.

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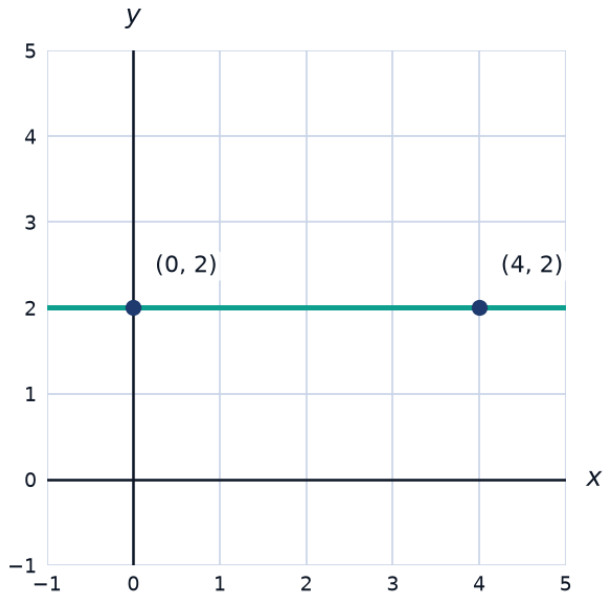
6 [Foundational] • Target: Total Distance vs. Displacement

The graph of the derivative of a function $f(x)$, denoted by $f'(x)$, is shown in the figure above. The function $f(x)$ is continuous, and it is known that $f(1) = 5$. Determine the value of $f(5)$.



7 [Foundational] • Target: Related Rates: Geometric Models

The graph of a function $f(x)$ is shown above. What is the instantaneous rate of change of $f(x)$ at $x = 2$? Show your complete algebraic work.

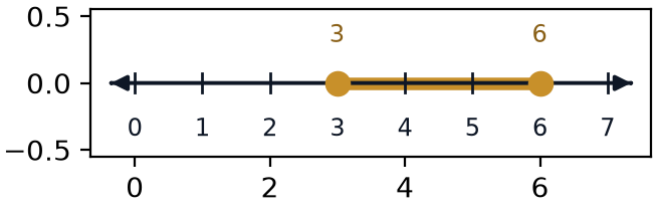


Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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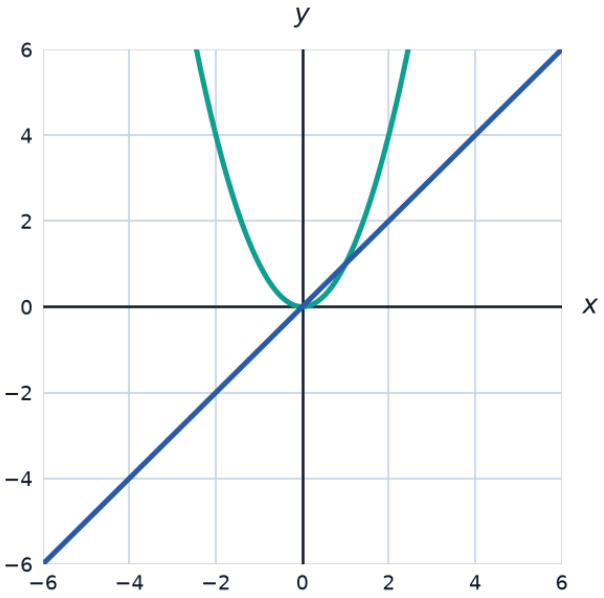
8 [Intermediate] • Target: Related Rates: Angles & Proportions

Consider the function $f(x) = 4$. What is the value of the definite integral $\int_3^6 f(x)\,dx$? Show your complete algebraic work.



9 [Intermediate] • Target: Position, Velocity & Speed

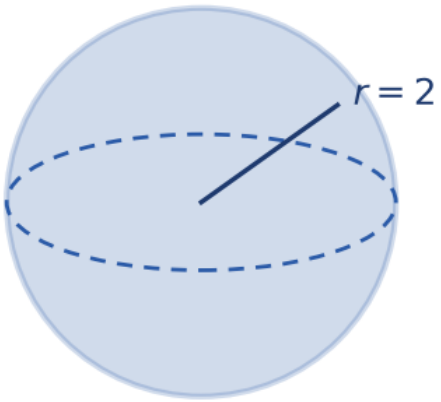
Area between $y = x^2$ and $y = x$ from 0 to 1 is:.



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10 **[Intermediate]** • **Target: Total Distance vs. Displacement**

Let R be the region bounded by the graph of $y = \sin(\pi x)$, the x-axis, and the lines $x = 0$ and $x = 1$. A solid is generated by revolving the region R about the x-axis. The radius of a cross-section of the solid perpendicular to the x-axis is given by $r(x) = k\sin(\pi x)$ for some positive constant k. If the volume of the solid is $\frac{\pi}{2}$ cubic units, determine the value of k.



11 **[Intermediate]** • **Target: Related Rates: Geometric Models**

The graph of the function $f(x) = ax - 3$ is shown below. The area of the shaded region between the graph of $f(x)$ and the x-axis from $x = 0$ to $x = 4$ is 20 square units. Find the value of the parameter a.

12 **[Intermediate]** • **Target: Related Rates: Angles & Proportions**

The velocity of a particle moving along a straight line is given by the function $v(t) = t^2 - 6t + 8$ m/s. What is the instantaneous acceleration of the particle at time $t = 3$ seconds? Show your complete algebraic work.

13 **[Intermediate]** • **Target: Position, Velocity & Speed**

A particle moves along a straight line so that its velocity at time $t \geq 0$ is modeled by $v(t) = 6t^2 - 18t + 12$. At what time t is the acceleration of the particle equal to 0? Show your complete algebraic work.

14 **[Intermediate]** • **Target: Total Distance vs. Displacement**

A differentiable function $f(x)$ is given by the equation $f(x) = x^2 - 6x + 10$. Find the instantaneous rate of change of $f(x)$ at $x = 4$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

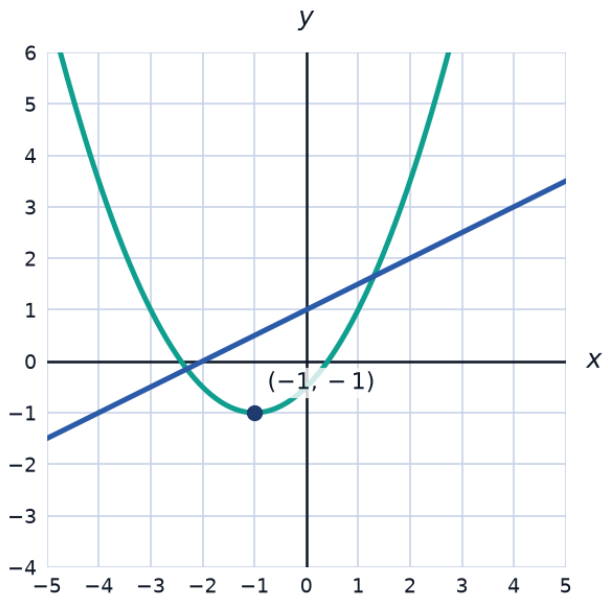
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15 [Challenging] • Target: Related Rates: Geometric Models • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Consider the function $f(x) = 3x^2 - 6x + 5$. What is the instantaneous rate of change of $f(x)$ at $x = 2$? Show your complete algebraic work.

16 [Challenging] • Target: Related Rates: Angles & Proportions • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

$\int e^x \, dx = .$



17 [Challenging] • Target: Position, Velocity & Speed • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

$\int_1^4 \frac{1}{x} \, dx = .$

18 [Challenging] • Target: Total Distance vs. Displacement • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

$\int 4x^3 \, dx = .$

19 [Challenging] • Target: Related Rates: Geometric Models • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

$\int_0^{\pi} \sin x \, dx = .$

20 [Challenging] • Target: Related Rates: Angles & Proportions • Cross-Lesson: 7.01 Differential Equations and Slope Fields

$\int_0^3 2x \, dx = .$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Position, Velocity & Speed

A particle moves along the x-axis with position $x(t) = t^3 - 6t^2 + 9t$ for $t \geq 0$. At what times is the particle at rest?

a. $t = 1$ and $t = 3$

b. $t = 0$ and $t = 3$

c. $t = 2$

d. $t = 3$ only

22 [Foundational] • Target: Position, Velocity & Speed

What is the speed of the particle when $v(t) = -8$ and $a(t) = 3$?

a. 8

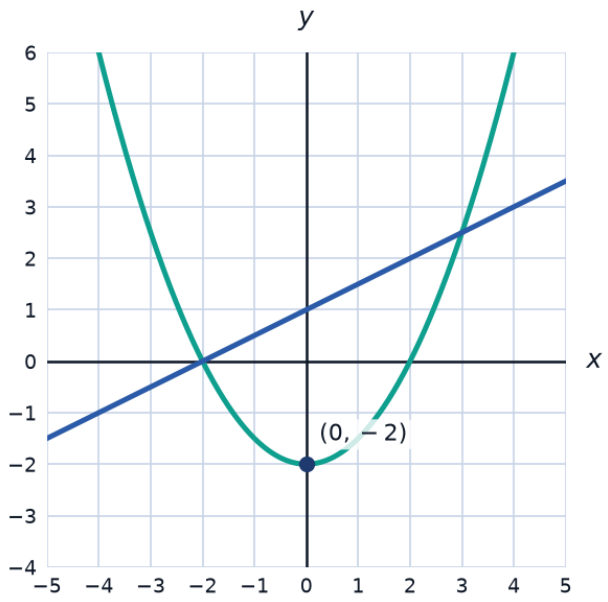
b. -8

c. 5

d. $\sqrt{73}$

23 [Foundational] • Target: Position, Velocity & Speed

When is a particle's speed increasing?



a. When velocity and acceleration have the same sign

b. When velocity is positive

c. When acceleration is positive

d. When acceleration is zero

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24 [Foundational] • Target: Total Distance vs. Displacement

If a particle's position is $s(t) = t^2 - 4t$ on $[0, 3]$, what is its net displacement?

a. -3

b. 5

c. -4

d. 3

25 [Foundational] • Target: Total Distance vs. Displacement

For the above particle with $s(t) = t^2 - 4t$ on $[0, 3]$, what is the total distance traveled?

a. 5

b. 3

c. 4

d. 7

26 [Foundational] • Target: Total Distance vs. Displacement

Which integral represents the total distance traveled by a particle from $t = a$ to $t = b$?

a. $\int_a^b |v(t)|dt$

b. $\int_a^b v(t)dt$

c. $|\int_a^b v(t)dt|$

d. $\int_a^b a(t)dt$

27 [Foundational] • Target: Related Rates: Geometric Models

The radius of a circle increases at a constant rate of 3 cm/s. How fast is the area increasing when the radius is 10 cm?

a. 60π cm^2/s

b. 30π cm^2/s

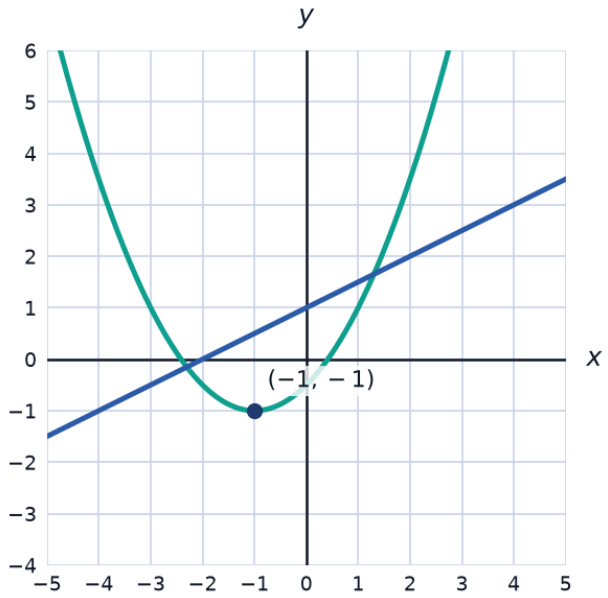
c. 100π cm^2/s

d. 20π cm^2/s

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28 [Foundational] • Target: Related Rates: Geometric Models

A 13-foot ladder leans against a vertical wall. If the bottom slides away from the wall at 2 ft/s, how fast is the top sliding down when the bottom is 5 feet from the wall?



a. $-\frac{5}{6}$ ft/s

b. $-\frac{12}{5}$ ft/s

c. -2 ft/s

d. $-\frac{5}{12}$ ft/s

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Related Rates: Geometric Models

Air is pumped into a spherical balloon at a rate of 100 cm^3/s. How fast is the radius increasing when $r = 5$ cm?

a. $\frac{1}{\pi}$ cm/s

b. $\frac{1}{4\pi}$ cm/s

c. $\frac{4}{\pi}$ cm/s

d. 25π cm/s

30 [Intermediate] • Target: Related Rates: Angles & Proportions

A conical tank with vertex down has height 10 m and base radius 4 m. Water flows in at 2 m^3/min. What is the relation between r and h?

a. $r = \frac{2}{5}h$

b. $r = \frac{5}{2}h$

c. $r = \frac{1}{2}h$

d. $r = 4h$

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31 [Intermediate] • Target: Related Rates: Angles & Proportions

Using $V = \frac{1}{3}\pi r^2 h = \frac{4}{75}\pi h^3$, how fast is the water level rising when $h = 5$ m and $\frac{dV}{dt} = 2$ m^3/min?

a. $\frac{1}{2\pi}$ m/min

b. $\frac{2}{\pi}$ m/min

c. $\frac{1}{4\pi}$ m/min

d. $\frac{4}{25\pi}$ m/min

32 [Intermediate] • Target: Related Rates: Angles & Proportions

A street light is mounted at the top of a 15-foot pole. A 6-foot man walks away from the pole at 5 ft/s. How fast is the tip of his shadow moving?

a. $\frac{25}{3}$ ft/s

b. 5 ft/s

c. $\frac{10}{3}$ ft/s

d. 10 ft/s

33 [Intermediate] • Target: Position, Velocity & Speed

A particle moves along the x-axis. Its velocity is given by the function $v(t) = \sqrt{t^2 + 1}$ for $t \geq 0$. The position of the particle at time $t = 0$ is $x(0) = 5$. Consider the function $P(x)$ defined as the total displacement of the particle from $t = 0$ to $t = x$. That is, $P(x) = x(x) - x(0)$. Which of the following statements about $P(x)$ is true for $x > 0$?

a. $P(x)$ is always positive and increasing.

b. $P(x)$ is always negative and decreasing.

c. $P(x)$ is always positive and decreasing.

d. $P(x)$ is always negative and increasing.

34 [Intermediate] • Target: Total Distance vs. Displacement

Given $\int_{-2}^{-1} 1x^2\,dx = \frac{7}{3}$ and $\int_{-1}^{0} 1x^2\,dx = \frac{1}{3}$, find $\int_{-2}^{0} 1x^2\,dx$.

a. $-\frac{8}{3}$

b. $\frac{1}{3}$

c. $\frac{8}{3}$

d. $\frac{7}{3}$

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35 [Intermediate] • Target: Related Rates: Geometric Models

Given $\int_{-1}^{1} 2x^2\,dx = \frac{4}{3}$ and $\int_{1}^{3} 2x^2\,dx = \frac{52}{3}$, find $\int_{-1}^{3} 2x^2\,dx$.

a. $\frac{4}{3}$

b. $\frac{52}{3}$

c. $-\frac{56}{3}$

d. $\frac{56}{3}$

36 [Intermediate] • Target: Related Rates: Angles & Proportions

Given $\int_{0}^{1} 2x^2\,dx = \frac{2}{3}$ and $\int_{1}^{2} 2x^2\,dx = \frac{14}{3}$, find $\int_{0}^{2} 2x^2\,dx$.

a. $\frac{14}{3}$

b. $\frac{2}{3}$

c. $\frac{16}{3}$

d. $-\frac{16}{3}$

37 [Intermediate] • Target: Position, Velocity & Speed

Given $\int_{-1}^{0} 4x^2\,dx = \frac{4}{3}$ and $\int_{0}^{1} 4x^2\,dx = \frac{4}{3}$, find $\int_{-1}^{1} 4x^2\,dx$.

a. $\frac{4}{3}$

b. $\frac{8}{3}$

c. $-\frac{8}{3}$

d. $\frac{17}{3}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Total Distance vs. Displacement • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Given $\int_{-2}^{1} 5x^2\,dx = 15$ and $\int_{1}^{5} 5x^2\,dx = \frac{620}{3}$, find $\int_{-2}^{5} 5x^2\,dx$.

a. $\frac{620}{3}$

b. 15

c. $-\frac{665}{3}$

d. $\frac{665}{3}$

39 [Challenging] • Target: Related Rates: Geometric Models • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Given $\int_{-2}^{1} 4x^2\,dx = 12$ and $\int_{1}^{3} 4x^2\,dx = \frac{104}{3}$, find $\int_{-2}^{3} 4x^2\,dx$.

a. 12

b. $\frac{104}{3}$

c. $-\frac{140}{3}$

d. $\frac{140}{3}$

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40 [Challenging] • Target: Related Rates: Angles & Proportions • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Given $\int_{-1}^{2} 4x^2\,dx = 12$ and $\int_{2}^{6} 4x^2\,dx = \frac{832}{3}$, find $\int_{-1}^{6} 4x^2\,dx$.

a. 12

b. $\frac{868}{3}$

c. $-\frac{868}{3}$

d. $\frac{832}{3}$

41 [Challenging] • Target: Position, Velocity & Speed • Cross-Lesson: 8.01 Area Between Curves

Given $\int_{-3}^{-1} 5x^2\,dx = \frac{130}{3}$ and $\int_{-1}^{0} 5x^2\,dx = \frac{5}{3}$, find $\int_{-3}^{0} 5x^2\,dx$.

a. $\frac{130}{3}$

b. -45

c. 45

d. $\frac{5}{3}$

42 [Challenging] • Target: Total Distance vs. Displacement • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Given $\int_{-2}^{0} 5x^2\,dx = \frac{40}{3}$ and $\int_{0}^{4} 5x^2\,dx = \frac{320}{3}$, find $\int_{-2}^{4} 5x^2\,dx$.

a. 120

b. $\frac{320}{3}$

c. -120

d. $\frac{40}{3}$

43 [Challenging] • Target: Related Rates: Geometric Models • Cross-Lesson: 1.01 Limits Graphically and Numerically

Given $\int_{-3}^{-2} 2x^2\,dx = \frac{38}{3}$ and $\int_{-2}^{0} 2x^2\,dx = \frac{16}{3}$, find $\int_{-3}^{0} 2x^2\,dx$.

a. $\frac{38}{3}$

b. -18

c. 18

d. $\frac{16}{3}$

44 [Challenging] • Target: Related Rates: Angles & Proportions • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Given $\int_{-1}^{2} 2x^2\,dx = 6$ and $\int_{2}^{6} 2x^2\,dx = \frac{416}{3}$, find $\int_{-1}^{6} 2x^2\,dx$.

a. $-\frac{434}{3}$

b. 6

c. $\frac{416}{3}$

d. $\frac{434}{3}$

45 [Challenging] • Target: Position, Velocity & Speed • Cross-Lesson: 2.01 The Definition of the Derivative

Given $\int_{-2}^{0} 4x^2\,dx = \frac{32}{3}$ and $\int_{0}^{3} 4x^2\,dx = 36$, find $\int_{-2}^{3} 4x^2\,dx$.

a. $\frac{140}{3}$

b. 36

c. $\frac{32}{3}$

d. $-\frac{140}{3}$

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